

Research Idea: J0604

ALMA Thesis Planner (automated pipeline)

Title: A First Characterization of H_2O^+ in $z = 2\text{--}5$ Starburst Sightlines: Detection Rate, Absorption Depth, and Velocity Extent from Program 2023.1.00804.S

One-sentence version. Using only the delivered ALMA measurement sets from program 2023.1.00804.S, the student characterizes the H_2O^+ feature itself — how often it appears, how deep it is, and how kinematically broad it is — in the $\sim 8\text{--}10$ sightlines with the strongest delivered continuum, and closes with one continuum-SNR-weighted stack of the full accessible sample for the ensemble-average profile.

The question. *How common is H_2O^+ absorption in the most active massive starbursts at $z = 2\text{--}5$, how deep is it, and how broad in velocity?* Nobody has a distribution for this species in this population; the program was built to get one. The thesis delivers the first version of that distribution from the data in hand and stops there.

The three measured quantities, per sightline. 1. *Classification.* Absorption, emission, P-Cygni, or non-detection, from the continuum-normalized profile against an explicitly stated significance threshold. Emission is a real possible outcome for a ground-state hydride against a bright continuum and is not treated as a failure mode. 2. *Integrated optical depth* over a fixed, generously wide velocity window, or a 3σ upper limit over that identical window for non-detections. Integrated τ is used rather than peak depth because in the optically thin limit it is conserved no matter how the transition’s oscillator strength is distributed among hyperfine components. 3. *Velocity extent* — Δv_{90} , the width of the contiguous interval containing 90% of the integrated τ . This is the one genuinely physical, genuinely robust number in the thesis: it is a *width*, not a shift, so it needs no systemic redshift; it is far less sensitive to hyperfine blending than a centroid; and because the hyperfine pattern is the same fixed pattern in every source, the blending contribution is common to all of them, so Δv_{90} *differences between sightlines* are meaningful even though each individual Δv_{90} is formally an upper bound on the intrinsic kinematic extent. That bound is stated every time the number is quoted.

Why $\sim 8\text{--}10$ targets, not all 40. Percent-level absorption is only measurable where the continuum is bright; a uniform analysis of the strongest-continuum subset is a complete result, whereas a nominally complete analysis of 40 sightlines is mostly upper limits computed at wildly unequal depth. The subset is chosen by delivered continuum SNR measured in step 1 below, with the ranking and the cut recorded, before any line measurement is looked at.

Headline deliverable: a detection rate with its sensitivity attached. The raw count of detections is meaningless without the depth each sightline could actually reach, so the detection rate is reported as a curve — the fraction of sightlines with detected H_2O^+ above a given integrated-optical-depth threshold, with the per-sightline 3σ limits shown alongside. Completeness is calibrated by injection–recovery: synthetic absorption of known integrated τ and known Δv_{90} is injected into the real normalized spectra and re-measured through the identical pipeline, which

both corrects the recovered fraction and quantifies how much Δv_{90} is biased by the measurement itself at each continuum brightness. No correlation is drawn against observed continuum flux density: across $z = 2-5$ that quantity mixes luminosity distance, the dust SED K-correction, dust temperature, and possible magnification, so a trend against it would carry no physical meaning. Continuum brightness enters this thesis only as what it is — the measurement’s sensitivity axis.

Line placement, with provenance handled honestly and no outside catalog. The data description quotes no frequencies, wavelengths, or redshifts; it gives only the redshift range $z = 2-5$ and the species (the “key oxygen hydrides,” H_2O^+ and OH^+). No numeric frequency is asserted here, and none is imported from an external spectroscopic database — the description names none, so none is available. The velocity axis is instead built entirely from what the data delivery itself records: the rest frequency carried in the delivered observing setup for each target’s H_2O^+ tuning, together with the per-target redshift recorded in that same delivered metadata. These are PI-supplied values; they are treated as approximate, the tabulated line frequency is labeled *rest-frame as recorded in the delivery* and every frequency derived from it by redshifting into the observed band is labeled *observed-frame and traceable to the recorded redshift*, and where the delivery does not state which frame a value is in, the thesis says so explicitly rather than picking one. This is tolerable precisely because every reported quantity is either an integral over a wide fixed window or a width, never a centroid or a velocity shift. Any sightline whose H_2O^+ region lands on a spectral-window edge or a strong atmospheric feature is excluded, with the reason recorded.

OH^+ is left as a question for the metadata to answer, not an assumption. The abstract describes the sample as *selected* for strong OH^+ signatures and the program as proposing to detect H_2O^+ , but whether OH^+ also falls inside a delivered spectral window is a fact the delivered observing setup states directly and that this proposal does not presume in either direction. The student reads it off the metadata in week one and records it. Nothing in the thesis depends on the answer: if OH^+ is in band for some sightlines, it is measured with the identical window-integrated procedure and reported as a parallel table in an appendix; if it is not, nothing is missing. No $N(\text{H}_2\text{O}^+)/N(\text{OH}^+)$ ratio is promised.

No chemical modeling. Converting hydride optical depths to abundances, molecular fraction, or fractional ionization requires a specific published steady-state network with its own rate coefficients, and the data description names none. The thesis reports integrated τ , equivalent width in km/s, Δv_{90} , and — under covering factor $C_f = 1$ as the fiducial, which makes every value a strict lower limit — a column density, with one alternative C_f shown to display the direction and size of the shift. Abundance and ionization modeling, and the program’s mass and momentum outflow/infall rates, are explicitly the follow-up work.

Scope, settled up front rather than branched. The acknowledgement text in the data description is the language used for data already in hand, so this thesis is scoped on the delivered measurement sets of 2023.1.00804.S as the group holds them. The one fact to confirm before a student is assigned — an afternoon’s work, not thesis work — is how many targets have delivered, QA2-passed data. That number sets the sample size, and the thesis is complete at any sample size down to a handful, because a detection rate with attached sensitivity limits and a depth/width distribution are real results at small N . There is no alternate thesis in reserve: the data description names no other dataset on these targets, and no substitute is proposed, because inventing one would misrepresent what is actually available.

Analysis, four steps. 1. *Continuum imaging and photometry.* Per target, per delivered spectral window: identify line-free channels, image with `tclean`, measure flux density and uncertainty at a

single documented aperture/fitting convention, and rank the sample by continuum SNR to fix the primary subset. 2. *Extraction and normalization.* Extract a spectrum at the continuum peak (and over the continuum-emitting aperture for any resolved source), divide by the fitted continuum, and propagate the *continuum-fit* uncertainty, not thermal noise alone, into the normalized profile. Establish the noise floor empirically by re-measuring blank velocity ranges in the same windows — baseline ripple, not raw sensitivity, is the realistic failure mode for percent-level absorption. 3. *Per-sightline characterization.* Classification, integrated τ or 3σ limit, equivalent width, Δv_{90} , and $C_f = 1$ column, all on the fixed window; injection–recovery run through the same code path for completeness and Δv_{90} bias. 4. *One stack.* Continuum-SNR-weighted stack of every accessible normalized spectrum on the metadata-defined velocity axis, giving the ensemble-average profile, mean depth, and mean Δv_{90} of the population — including the sightlines too faint for individual measurement.

Deliverables. (a) The detection-rate-versus-depth-threshold result with per-sightline 3σ limits and injection–recovery completeness — the headline; (b) the depth and Δv_{90} distributions across the strong-continuum subset, with the hyperfine upper-bound caveat stated inline; (c) a spectral atlas of every analyzed sightline’s continuum-normalized H_2O^+ region on a common velocity axis, with classifications; (d) the single full-sample stacked profile; (e) *secondary, stated as secondary:* the uniform continuum photometry table produced in step 1, useful to the group but not a science claim of this thesis; (f) *appendix, contingent on the metadata:* the identical OH^+ measurements for any sightlines where OH^+ is in band.

Why it can conclude something. A measured detection fraction, a depth distribution, and a velocity-extent distribution for H_2O^+ in $z = 2\text{--}5$ starburst sightlines are the empirical facts the program’s whole science case is built on, and none of them currently exists. If H_2O^+ is common and broad, that is a result; if it is rare or narrow, that is a result and a real constraint on the ionized-hydride absorbing layers in early-universe starbursts; if the stack is a non-detection, the stacked limit is itself the first quantitative ceiling on the species in this population. Reduction scripts are working notes, not a graded software deliverable.